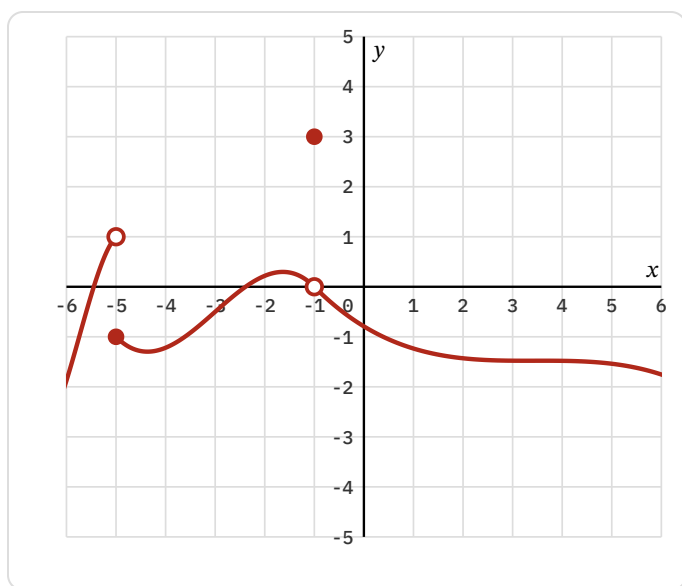


Mock Midterm B

Practice midterm (AI-Free) · Sections 1.1–1.4

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-5)$ _____ (1)
- b.) $\lim_{x \rightarrow -5^-} f(x)$ _____ (1)
- c.) $\lim_{x \rightarrow -5^+} f(x)$ _____ (1)
- d.) $\lim_{x \rightarrow -5} f(x)$ _____ (1)
- e.) $f(-1)$ _____ (1)
- f.) $\lim_{x \rightarrow -1^-} f(x)$ _____ (1)
- g.) $\lim_{x \rightarrow -1^+} f(x)$ _____ (1)
- h.) $\lim_{x \rightarrow -1} f(x)$ _____ (1)

2. Let $f(x) = \frac{\sqrt{x}-3}{x-9}$. We want to know what happens to the outputs when x is near **9**.

7 pts

x	8.9	8.99	8.999	9	9.001	9.01	9.1
$f(x)$	0.167132	0.166713	0.166671	?	0.166662	0.16662	0.166206

a.) Can you find $\lim_{x \rightarrow 9} f(x)$ by plugging in $x = 9$? (1)

(A) Yes — just compute $f(9)$

(B) No — substituting gives $\frac{0}{0}$, which is undefined

(C) No — f is undefined for every x near **9**

b.) As $x \rightarrow 9^-$, $f(x)$ approaches... (2)

c.) As $x \rightarrow 9^+$, $f(x)$ approaches... (2)

d.) $\lim_{x \rightarrow 9} f(x) =$ (2)

3. Calculate each limit algebraically.

6 pts

a.) $\lim_{x \rightarrow -4} (2x^2 + 3x + 2)$ (2)

b.) $\lim_{x \rightarrow -3} \frac{x^2 - x - 9}{x - 2}$ (2)

c.) $\lim_{x \rightarrow 2} \sqrt{x^2 + 21}$ (2)

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4}$ _____ (2)

b.) $\lim_{h \rightarrow 0} \frac{(2+h)^3-8}{h}$ _____ (2)

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results. 6 pts

a.) $\lim_{x \rightarrow -\infty} \frac{4x^2+1}{6x^2-9}$ _____ (2)

b.) $\lim_{x \rightarrow \infty} \frac{4}{-x^4-7x^2}$ _____ (2)

c.) $\lim_{x \rightarrow \infty} \frac{x^5-7x^4+7x^3-3x^2+7x+4}{-4x^3+9x^2}$ _____ (2)

6. Let

7 pts

$$G(x) = \begin{cases} -x^2 & x < -1 \\ 2x + 1 & x > -1 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(-1)$ _____ (1)

b.) $\lim_{x \rightarrow -1^-} G(x)$ _____ (1)

c.) $\lim_{x \rightarrow -1^+} G(x)$ _____ (1)

d.) $\lim_{x \rightarrow -1} G(x)$ _____ (1)

e.) $G(2)$ _____ (1)

f.) $\lim_{x \rightarrow -\infty} G(x)$ _____ (1)

g.) $\lim_{x \rightarrow \infty} G(x)$ _____ (1)

7. Let

6 pts

$$f(x) = \begin{cases} \frac{x^2 - 2x - 8}{x + 2} & x < -2 \\ 3x + 2 & x \geq -2 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 0} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -4} f(x)$ _____ (1)

c.) $f(-2)$ _____ (1)

d.) $\lim_{x \rightarrow -2^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow -2^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow -2} f(x)$ _____ (1)

8. Find the value of k that makes f continuous at $x = -2$.

4 pts

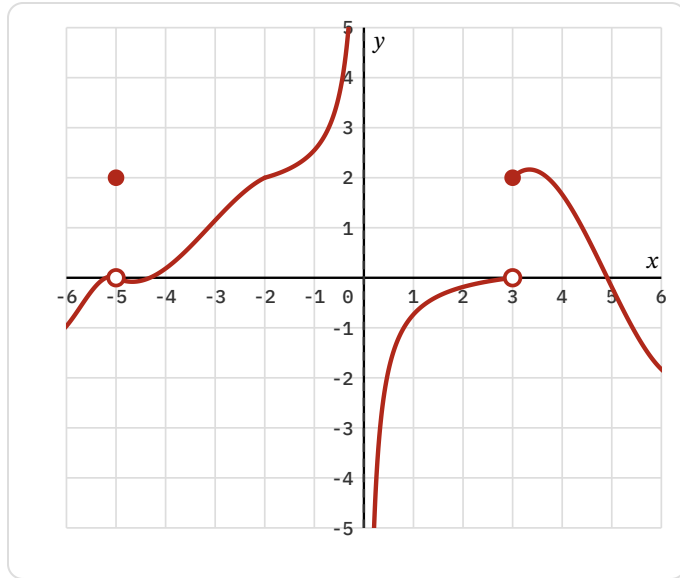
$$f(x) = \begin{cases} x^2 + x & x < -2 \\ kx + 6 & x \geq -2 \end{cases}$$

a.) $\lim_{x \rightarrow -2^-} f(x)$ _____ (2)

b.) $k =$ _____ (2)

9. Classify the behavior of f at each labeled x -value.

4 pts



a.) At $x = -5$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

b.) At $x = -2$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

c.) At $x = 0$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

d.) At $x = 3$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

10. Let $C(q) = 3q^2 + 2q + 163$, where $C(q)$ represents the cost, in dollars, of producing q items. 7 pts
Find the average rate of change over $[1, 5]$.

a.) $C(1)$ _____ (2)

b.) $C(5)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

d.) Which sentence correctly states the result **with units**? (1)

(A) Between $q = 1$ and $q = 5$ items, the cost changes at an average rate of **20** dollars per item.

(B) The cost changes by a total of **20** dollars from $q = 1$ to $q = 5$.

(C) At exactly $q = 5$ items, the cost is changing at a rate of **20** dollars per item.

(D) Between $q = 1$ and $q = 5$, the cost changes at an average rate of **20** items per dollar.

11. Let $f(x) = \frac{6}{x}$. Calculate the average rate of change of f over each interval (rounded answers to 2+ decimals are fine). 5 pts

a.) $[2, 5]$ _____ (1)

b.) $[2, 3]$ _____ (1)

c.) $[2, 2.5]$ _____ (1)

d.) $[2, 2.05]$ _____ (1)

e.) As the interval shrinks toward $x = 2$, these values approach... (1)

(A) $-\frac{3}{2}$

(B) $\frac{3}{2}$

(C) 0

(D) -3

12. Let $f(x) = x^3 - 3x$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table. 7 pts

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) $x = -1, h = 1$ _____ (1)

c.) $x = -1, h = 0.1$ _____ (1)

d.) $x = -1, h = 0.01$ _____ (1)

e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

13. Let $f(x) = 3x^2 - 2x - 6$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ _____ (3)

b.) $f'(-3) =$ _____ (2)

c.) Slope of the tangent line to f at $x = 2$ _____ (2)

d.) What does $f'(-3) = -20$ mean? (1)

(A) The slope of the secant line from $x = -3$ to $x = -3 + 1$ is -20 .

(B) The value of f at $x = -3$ is -20 .

(C) The average rate of change of f from $x = 0$ to $x = -3$ is -20 .

(D) The instantaneous rate of change of f at $x = -3$ is -20 — the slope of the tangent line there.

14. Let $f(x) = x^2 - 8x - 3$, so $f'(x) = 2x - 8$.

6 pts

- a.) At what x is the tangent line horizontal? _____ (2)
- b.) What is the y -coordinate of that point? _____ (2)
- c.) At what x does the tangent line have slope 2? _____ (2)

15. Let $h(t) = -16t^2 + 128t$ be the height, in feet, of a rocket t seconds after launch.

6 pts

- a.) $h'(t) =$ _____ (3)
- b.) $h'(5)$ _____ (2)
- c.) Interpret $h'(5) = -32$. (1)

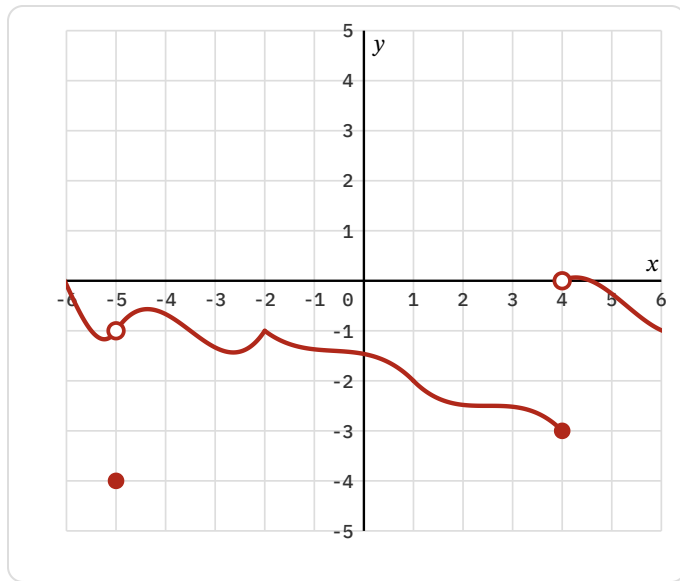
(A) Over the first 5 seconds, the height of the rocket changed at an average rate of 32 feet per second.

(B) At $t = 5$ seconds, the height of the rocket is 32 feet.

(C) At $t = 5$ seconds, the height of the rocket is decreasing at a rate of 32 feet per second.

(D) At $t = 5$ seconds, the height of the rocket is increasing at a rate of 32 feet per second.

16. At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there. 4 pts



- a.) At $x = -5$ (1)
- (A) Continuous and differentiable
 - (B) Continuous but NOT differentiable
 - (C) Not continuous (so not differentiable)
- b.) At $x = -2$ (1)
- (A) Continuous and differentiable
 - (B) Continuous but NOT differentiable
 - (C) Not continuous (so not differentiable)
- c.) At $x = 1$ (1)
- (A) Continuous and differentiable
 - (B) Continuous but NOT differentiable
 - (C) Not continuous (so not differentiable)
- d.) At $x = 4$ (1)
- (A) Continuous and differentiable
 - (B) Continuous but NOT differentiable
 - (C) Not continuous (so not differentiable)